NANYANG TECHNOLOGICAL UNIVERSITY

SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 1 MH1100 Calculus I Tutorial 2, Week 3

Your tutor will aim to discuss: Problem 3, 6, 8, 9, and 10

Problem 1 The point $P(1,1)$ lies on the the graph of the function $f(x) = \frac{3x}{1+2x}$.

(a) If Q is the point $\left(x, \frac{3x}{1+2x}\right)$, use your calculator to find the slope of the secant line PQ for the values of x: (i) 0.5; (ii) 0.9; (iii) 0.99; (iv) 0.999; (v) 1.5; (vi) 1.1; (vii) 1.01; and (viii) 1.001.

(b) Using the results of part (a), guess the value of the slope of the tangent line to the curve at $P(1,1)$.

(c) Using the slope from part (b), find an equation of the tangent line to the curve at $P(1,1)$.

[Solution:]

(a) In this problem we are asked to guess the slope of the tangent to the graph of the function $f(x) = \frac{3x}{1+2x}$ at the point $P(1,1)$ by looking at the slopes of various secants starting at that point. Consider, for example, the first case we are asked to use: the secant from $P(1,1)$ to $Q(0.5, f(0.5))$. Here is a graph of that secant (in red):

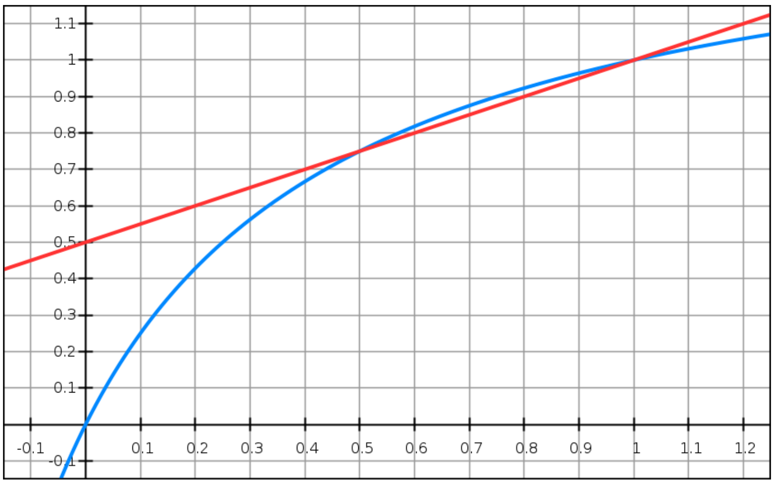

FIGURE 1

The slope of this secant is given by the calculation:

$$m_{0.5} = \frac{f(1.0) - f(0.5)}{1.0 - 0.5} = \frac{1 - \frac{1.5}{2}}{1.0 - 0.5} = \frac{1}{2}.$$

The other slopes are as follows.

$$m_{0.9} = \frac{f(1.0) - f(0.9)}{1.0 - 0.9} = \frac{1 - \frac{2.7}{2.8}}{1.0 - 0.9} = \frac{10}{28}.$$

$$m_{0.99} = \frac{f(1.0) - f(0.99)}{1.0 - 0.99} = \frac{1 - \frac{2.97}{2.98}}{1.0 - 0.99} = \frac{100}{298}.$$

$$m_{0.999} = \frac{f(1.0) - f(0.999)}{1.0 - 0.999} = \frac{1 - \frac{2.997}{2.998}}{1.0 - 0.999} = \frac{1000}{2998}.$$

$$m_{1.5} = \frac{f(1.0) - f(1.5)}{1.0 - 1.5} = \frac{1 - \frac{4.5}{4.0}}{1.0 - 1.5} = \frac{1}{4}.$$

$$m_{1.1} = \frac{f(1.0) - f(1.1)}{1.0 - 1.1} = \frac{1 - \frac{3.3}{3.2}}{1.0 - 1.1} = \frac{10}{32}.$$

$$m_{1.01} = \frac{f(1.0) - f(1.01)}{1.0 - 1.01} = \frac{1 - \frac{3.03}{3.02}}{1.0 - 1.01} = \frac{100}{302}.$$

$$m_{1.001} = \frac{f(1.0) - f(1.001)}{1.0 - 1.001} = \frac{1 - \frac{3.003}{3.002}}{1.0 - 1.001} = \frac{1000}{3002}.$$

(b) Considering these slopes, we can say that the slopes seem to be approaching the value 1/3 as x approaches 1 from the left and also from the right. So our guess for the slope of the tangent to the graph of $f(x)$ at $x = 1$ is 1/3.

(c) The tangent, then, is the straight line going through the point $(1, 1)$ with gradient 1/3. That line is given by the equation:

$$y - 1 = \frac{1}{3}(x - 1) \qquad \text{or} \qquad y = \frac{1}{3}x + \frac{2}{3}.$$

Here is a plot of that tangent, which suggests that our guess is a good approximation.

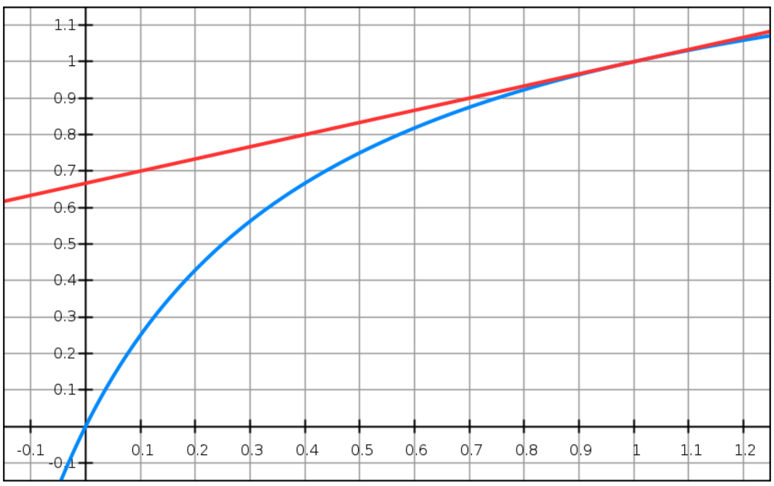

Figure 2

Problem 2 The displacement of a particle moving back and forth along a straight line is given by the equation of motion $s = 2\sin \pi t + 3\cos \pi t$, where t is measured in seconds.

(a) Find the average velocity for during the time period : (i) $[1, 2]$; (ii) $[1, 1.1]$; (iii) $[1, 1.01]$; (iv) $[1, 1.001]$.

(b) Estimate the instantaneous velocity of the particle when $t = 1.0$.

[Solution:]

(a) The average velocity over the brief time interval of τ starting from t to $t + \tau$ can be computed in the following way.

$$v_\tau(t) = \frac{s(t+\tau) - s(t)}{\tau}.$$

So we have

$$\begin{aligned} v_{1.0}(1.0) &= \frac{s(1.0+1.0) - s(1.0)}{1.0} \\ &= \frac{(2\sin 2\pi + 3\cos 2\pi) - (2\sin \pi + 3\cos \pi)}{1.0} \\ &= \frac{3 - (-3)}{1.0} = 6 \end{aligned}$$

$$\begin{aligned} v_{0.1}(1.0) &= \frac{s(1.0+0.1) - s(1.0)}{0.1} \\ &= \frac{(2\sin 1.1\pi + 3\cos 1.1\pi) - (2\sin \pi + 3\cos \pi)}{0.1} \\ &\approx -4.7120 \end{aligned}$$

$$\begin{aligned} v_{0.01}(1.0) &= \frac{s(1.0+0.01) - s(1.0)}{0.01} \\ &= \frac{(2\sin 1.01\pi + 3\cos 1.01\pi) - (2\sin \pi + 3\cos \pi)}{0.01} \\ &\approx -6.1341 \end{aligned}$$

$$\begin{aligned} v_{0.001}(1.0) &= \frac{s(1.0+0.001) - s(1.0)}{0.001} \\ &= \frac{(2\sin 1.001\pi + 3\cos 1.001\pi) - (2\sin \pi + 3\cos \pi)}{0.001} \\ &\approx -6.2685 \end{aligned}$$

(b) We were asked to estimate the instantaneous velocity when $t = 1.0$. Our calculations in Part(a) cannot lead us to any estimate. But the instantaneous velocity is actually $-2\pi \approx -6.28318$.

Problem 3 Consider a function $h(x)$ with the graph as shown in Figure 3. State the value of the following quantity, if it exists. If it does not exist, explain why.

(a) $\lim_{x\to -3^-} h(x)$	(b) $\lim_{x\to -3^+} h(x)$	(c) $\lim_{x\to -3} h(x)$	(d) $h(-3)$
(e) $\lim_{x\to 0^-} h(x)$	(f) $\lim_{x\to 0^+} h(x)$	(g) $\lim_{x\to 0} h(x)$	(h) $h(0)$
(i) $\lim_{x\to 2} h(x)$	(j) $h(2)$	(k) $\lim_{x\to 5^+} h(x)$	(l) $\lim_{x\to 5^-} h(x)$

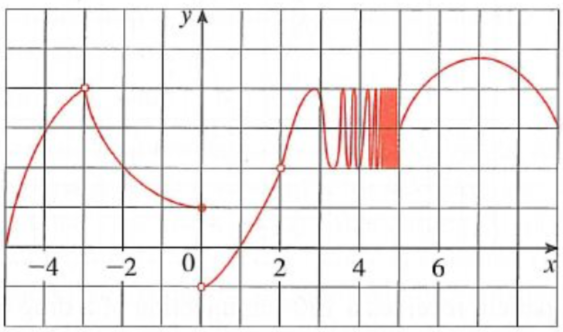

FIGURE 3. Graph of $h(x)$ in Problem 3

[Solution:] We were given the graph of some function $h(x)$, and are asked to determine a number of quantities, or explain why they do not exist. We find:

(a) $\lim_{x\to -3^-} h(x) = 4$.

(b) $\lim_{x\to -3^+} h(x) = 4$.

(c) $\lim_{x\to -3} h(x) = 4$. Comment: The two 1-sided limits exist and are equal, so the limit exists, even though the function is defined by two different rules on the two sides of -3, and $h(x)$ is not even defined at that point.

(d) $h(-3)$ does not exist. -3 is not in the domain of $h(x)$.

(e) $\lim_{x\to 0^-} h(x) = 1$.

(f) $\lim_{x\to 0^+} h(x) = -1$.

(g) $\lim_{x\to 0} h(x)$ does not exist. Comment: The two 1-sided limits both exist, but are different, so this limit does not exist.

(h) $h(0) = 1$.

(i) $\lim_{x\to 2} h(x) = 2$.

(j) $h(2)$ does not exist. Comment: again, note that even though the function is not defined at 2, the limit at that point exists.

(k) $\lim_{x\to 5^+} h(x) = 3$.

(l) $\lim_{x\to 5^-} h(x) = 3$ does not exist. Comment: As we approach 5 from the left, the function does not seem to approach any particular value. It just keeps oscillating.

Problem 4 Give an example of a function f satisfying the following conditions:
$\lim_{x\to 0^-} f(x) = 1$, $\lim_{x\to 0^+} f(x) = -1$, $\lim_{x\to 2^-} f(x) = 0$, $\lim_{x\to 2^+} f(x) = 1$, $f(2) = 1$, and $f(0)$ is undefined.

[Solution:] This is an example of a problem which can be solved in many, in fact, countless, different ways. For example, here are two equally correct solutions:

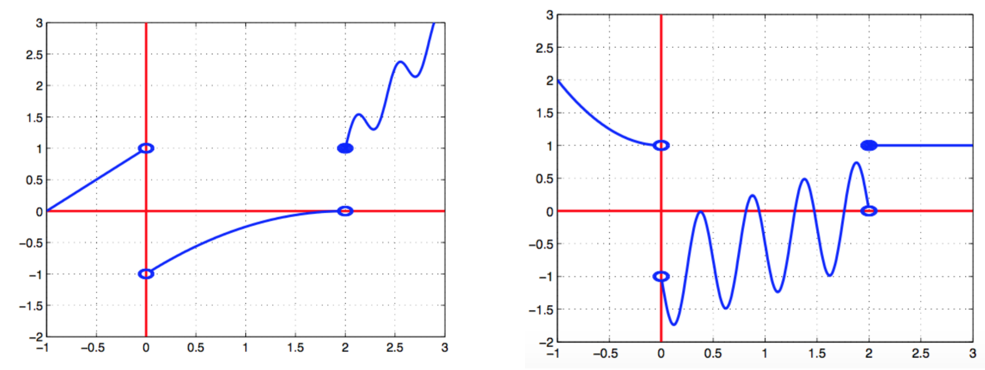

FIGURE 4

Problem 5 Guess the value of the limit

$$\lim_{x \to 3} \frac{x^2 - 3x}{x^2 - 2x - 3}$$

by evaluating the function at the points

$$x = 3.5, 3.1, 3.05, 3.01, 3.005, 2.5, 2.9, 2.95, 2.99, 2.995.$$

[Solution:] This problem asks us to guess the limit:

$$\lim_{x \to 3} \frac{x^2 - 3x}{x^2 - 2x - 3}$$

by evaluating the function $f(x) = \frac{x^2-3x}{x^2-2x-3}$ at a whole bunch of different points. Using a calculator, we get the values:

x	3.5	3.1	3.05	3.01	3.005	2.5	2.9	2.95	2.99	2.995
f(x)	0.7778	0.7561	0.7531	0.7506	0.7503	0.7143	0.7436	0.7468	0.7494	0.7497

These values lead us to guess that the limit is equal to

$$\lim_{x \to 3} \frac{x^2 - 3x}{x^2 - 2x - 3} = 0.75.$$

Problem 6

(a) Evaluate the function $f(x) = x^2 - (2^x/1000)$ for $x = 1$, 0.8, 0.6, 0.4, 0.2, 0.1, and 0.05, and guess the value of

$$\lim_{x \to 0} \left(x^2 - \frac{2^x}{1000} \right).$$

(b) Evaluate $f(x)$ for $x = 0.04$, 0.02, 0.01, 0.005, 0.003, and 0.001. Guess again.

[Solution:]

(a) Using a calculator, we get the values:

x	1.0	0.8	0.6	0.4	0.2	0.1	0.05
f(x)	0.998000	0.638259	0.358484	0.158680	0.038851	0.008928	-0.000978

It appears that

$$\lim_{x\to 0} f(x) = 0.$$

(b) We evaluate the values of f for $x = 0.04,\ 0.02,\ 0.01,\ 0.005,\ 0.003,$ and 0.001 as the below:

x	0.04	0.02	0.01	0.005	0.003	0.001
f(x)	0.000572	-0.000614	-0.000907	-0.000978	-0.000993	-0.0009997

We may guess that

$$\lim_{x\to 0} f(x) = -0.001.$$

Problem 7 Consider the function $f(x) = \frac{\sin \pi x}{\sin x}$.

(a) Estimate the value of $\lim\limits_{x\to 0} f(x)$ by graphing the function $f(x)$. State your answer correct to two decimal places.

(b) Check your answer in part(a) by evaluating $f(x)$ for values of x that approach 0.

[Solution:]

(a) We sketch the graph of the function $f(x)$ on the interval $(-0.04, 0.04) \setminus \{0\}$. From the graph below, we can observe that when x is around 0, the values of $f(x)$ are in between 3.140 and 3.145. For the answer correct to two decimal places, we can guess that

$$\lim_{x\to 0} \frac{\sin \pi x}{\sin x} = 3.14.$$

(b) We can check that $f(x)$ is an even function in its domain $(-\infty, \infty) \setminus \{0\}$. So we only need to check the values of f as x approaches 0 from the right. Using a calculator, we get the values of f at 6 points close to 0:

x	0.005	0.004	0.003	0.002	0.001	0.0005
f(x)	3.141477	3.141518	3.141551	3.141574	3.141588	3.141592

This is consistent with the result in Part(a).

Problem 8 Consider the function $f(x) = \tan \frac{\pi}{x}$.

(a) Show that $f(x) = 0$ for $x = \frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \cdots, \frac{1}{n}, \cdots$.

(b) Show that $f(x) = 1$ for $x = \frac{4}{1}, \frac{4}{5}, \frac{4}{9}, \cdots, \frac{4}{4n-3}, \cdots$.

(c) What can you conclude about $\lim\limits_{x\to 0^+} \tan \frac{\pi}{x}$?

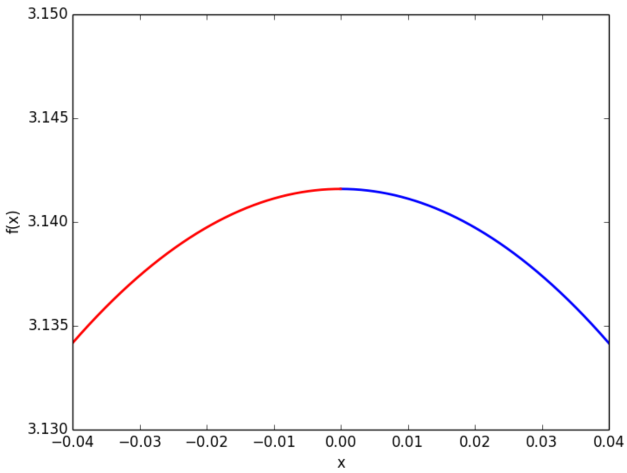

FIGURE 5

[Solution:]

(a) For any positive integer n, if $x = \frac{1}{n}$, then $f(x) = \tan\frac{\pi}{x} = \tan(n\pi) = 0$. (Remember that the tangent function has period π.)

(b) For any positive integer n, if $x = \frac{4}{4n-3}$, then

$$\begin{aligned} f(x) &= \tan\frac{\pi}{x} = \tan\frac{(4n-3)\pi}{4} \\ &= \tan\frac{(4n-4+1)\pi}{4} = \tan\left[(n-1)\pi + \frac{\pi}{4}\right] \\ &= \tan\frac{\pi}{4} = 1. \end{aligned}$$

(c) From part (a), $f(x) = 0$ infinitely often as x approaches 0. From part (b), $f(x) = 1$ infinitely often as x approaches 0. Thus, $\lim_{x\to 0^+} \tan\frac{\pi}{x}$ does not exist since $f(x)$ does not get close to a fixed number as x approaches 0.

Problem 9

(a) Use numerical and graphical evidence to guess the value of the limit

$$\lim_{x\to 4} \frac{x^3 - 64}{\sqrt{x} - 2}.$$

(b) How close to 4 does x have to be to ensure that the function in part (a) is within a distance 2.0 of its limit?

[Solution:]

(a) We evaluate the values of f at different points close to $x = 4$.

x	3.9	3.99	3.999	4.001	4.01	4.1
f(x)	186.0623	191.4006	191.9400	192.0600	192.6006	198.0627

We also draw the graph of the function $\frac{x^3-64}{\sqrt{x}-2}$ around $x = 4$. From the table and the

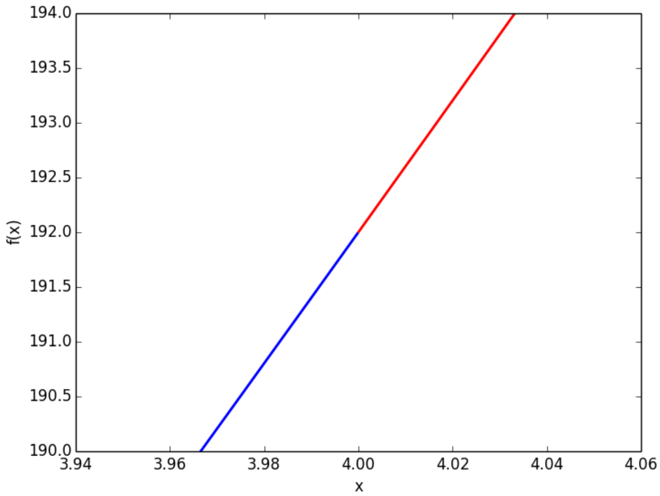

FIGURE 6. Graph of $y = \frac{x^3-64}{\sqrt{x}-2}$ in Problem 12.

graph, we guess that the limit of y as x approaches 4 is 192.

(b) We need to have $190.0 < \frac{x^3-64}{\sqrt{x}-4} < 194.0$. From the graph we obtain the approximate points of intersection $(3.967, 190.0)$ and $(4.033, 194.0)$. Now $4 - 3.967 = 0.033$ and $4.033 - 4.0 = 0.033$, so by requiring that x be within 0.033 of 4, we may have that y is within 2.0 of 192. To make sure that y is within 2.0 of 192, we can require that x be within 0.02 of 4. As shown in the graph, we can observe that when $3.98 < x < 4.02$, $190.0 < y < 194.0$.

Problem 10 Let $f(t) = 1/t$ for $t \neq 0$.

(a) Find the average rate of change of f with respect to t over the intervals (i) from $t = 2$ to $t = 3$, and (ii) from $t = 2$ to $t = T$.

(b) Make a table of values of the average rate of change of f with respect to t over the interval $[2, T]$, for some values of T approaching 2, say $T = 2.1$, 2.01, 2.001, 2.0001, 2.00001, and 2.000001.

(c) What does your table indicate is the rate of change of f with respect to t at $t = 2$?

(d) Calculate the limit as T approaches 2 of the average rate of change of f with respect to t over the interval from 2 to T. You will have to do some algebra before you can substitute $T = 2$.

[Solution:]

(a) **The average rate of change of f with respect to t over the interval $[t_1, t_2]$ is**

$$\frac{f(t_2) - f(t_1)}{t_2 - t_1}.$$

Thus, over the interval of [2, 3], the average rate of change of f with respect to t is

$$\frac{f(3) - f(2)}{3 - 2} = \frac{\frac{1}{3} - \frac{1}{2}}{3 - 2} = -\frac{1}{6}.$$

The average rate of change of f with respect to t over the interval from $t = 2$ to $t = T$ is

$$\frac{f(T) - f(2)}{T - 2} = \frac{\frac{1}{T} - \frac{1}{2}}{T - 2} = \frac{\frac{2-T}{2T}}{T - 2} = -\frac{1}{2T}.$$

(b)

T	2.1	2.01	2.001	2.0001	2.00001	2.000001
$\frac{f(T)-f(2)}{T-2} = -\frac{1}{2T}$	-0.243902	-0.248756	-0.249875	-0.249986	-0.249999	-0.250000

(c) From the table we guess that the rate of change of f with respect to t at $t = 2$ is -0.25.

(d)

$$\lim_{T \to 2} \frac{f(T) - f(2)}{T - 2} = \lim_{T \to 2} \left(-\frac{1}{2T} \right) = -\frac{1}{4}.$$